

Reviewer Pack — Minimal Order of Symplectic Leaves vs. DPLL Search Tree Width...

Entry 18383e398d8f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Symplectic Leaves vs. DPLL Search Tree Width

Entry ID: 18383e398d8f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 17:54:50 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Complexity Theory: DPLL Search Tree Complexity

Statement:

[‘For every n -vertex symmetric Boolean circuit C with DPLL search tree width $w(C)$, there exists a symplectic leaf l in the symplectic leaf space of an n -dimensional symplectic vector space such that the minimal order of the corresponding divisor on the projective Grassmannian is $O(w(C))$.’, ‘Equivalently, the number of distinct symplectic leaves contributing to the divisor is upper bounded by a function $f(n) = \Theta(w(C))$.’]

Rationale (proposer’s reasoning):

[‘Symplectic geometry has not been widely applied to complexity theory, especially in the context of DPLL search tree width. This conjecture proposes a potential connection that could reveal new structural insights into computational complexity.’, ‘If true, it would suggest that symplectic structures can be used to bound computational complexity in a meaningful way, potentially leading to new algorithms or proving techniques.’]

Taxonomy category: SYMPLECTIC_GEO_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e5db230e7cbbb25b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 30 random seeds, the minimal order of the divisor on the projective Grassmannian associated with symplectic leaves from n -vertex symmetric Boolean circuits matches $O(w(C))$ with an aggregate mean difference between the measured minimal order and $w(C) \leq 1.5$ across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "symplectic geometry" AND "DPLL search tree width" - "projective Grassmannian" AND minimal order symplectic leaves AND DPLL complexity" - " $\theta(w(C))$ " AND symplectic geometry AND DPLL search tree

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.2s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_symmetric_boolean_circuit(n):
        circuit = []
        for _ in range(n):
            gate_type = random.choice(['AND', 'OR'])
            inputs = [random.randint(0, 1) for _ in range(random.randint(2, n))]
            circuit.append((gate_type, inputs))
        return circuit

    def dpll_search_tree_width(circuit):
        # Simplified DPLL width calculation
        max_width = 0
        current_width = 0
        stack = []
        for gate in circuit:
            if gate[0] == 'AND':
                current_width += len(gate[1])
                stack.append(current_width)
                max_width = max(max_width, current_width)
            elif gate[0] == 'OR':
                current_width -= len(stack.pop())
        return max_width

    def symplectic_leaf_space(circuit):
        # Simplified symplectic leaf space calculation
```

```

    leaves = set()
    for gate in circuit:
        if gate[0] == 'AND':
            leaves.add(tuple(gate[1]))
        elif gate[0] == 'OR':
            leaves.remove(tuple(gate[1]))
    return leaves

def minimal_order_of_divisor(leaves):
    # Simplified minimal order calculation
    return len(leaves)

n = random.randint(5, 40)
circuit = generate_symmetric_boolean_circuit(n)
w_C = dp11_search_tree_width(circuit)
leaves = symplectic_leaf_space(circuit)
min_order = minimal_order_of_divisor(leaves)

return {
    "metric_name": "minimal_order",
    "metric_value": min_order,
    "instances_tested": 1,
    "conjecture_holds": abs(min_order - w_C) <= 1.5,
    "counterexample": "" if conjecture_holds else f"n={n}, w(C)={w_C}, min_order={min_order}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30)) # Default to first 30 primes if

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        counterexample = next(r["counterexample"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b173887a.py", line 77, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b173887a.py", line 59, in run_trial
    w_C = dpll_search_tree_width(circuit)
              ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b173887a.py", line 40, in dpll_search_t
    current_width -= len(stack.pop())
                      ^^^^^^^^^^^^^
IndexError: pop from empty list
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete its intended task of verifying the conjecture. | next: Re-run the test with proper error handling to ensure that it can complete without crashing and provide the necessary data for verification.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14508
2	preregistration	ollama_remote	glm4:latest	0	0	9411
3	novelty	ollama_remote	glm4:latest	0	0	8698
4	novelty	ollama_remote	glm4:latest	0	0	8834
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11776
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6739
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19562
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9188
9	judge	ollama_remote	glm4:latest	0	0	24707

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 113423 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/18383e398d8f.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice

3. The full LLM transcripts are in `research/audit/18383e398d8f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/18383e398d8f.tar.gz` (if generated)